

Midterm Practice Solutions

Problem 1

1.

- population: all products produced by the machine.
- sample: selected products produced by the machine.
- estimand: the expected time needed for the machine to produce a product.

2.

```
for (i in 1:n) {  
  draw y_i from Expo(\lambda)  
}
```

3.

The likelihood function:

$$L(\lambda; y_1, \dots, y_n) = \lambda^n e^{-\lambda \sum_{i=1}^n y_i}$$

Taking log of the likelihood function to get the log-likelihood:

$$l(\lambda; y_1, \dots, y_n) = n \log \lambda - \lambda \sum_{i=1}^n y_i$$

4.

Taking derivative of log-likelihood function:

$$l'(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^n y_i$$

Setting $l'(\lambda) = 0$ and solve it for λ :

$$\hat{\lambda}_{MLE} = \frac{n}{\sum_{i=1}^n y_i}$$

5.

Gamma. This can be demonstrated by showing that both prior and posterior can be written in the form $\lambda^{\alpha-1} e^{-\beta\theta}$ for appropriate values of α and β

6.

$$\begin{aligned} p(\lambda \mid y_1, \dots, y_n) &\propto \lambda^n e^{-\lambda \sum_{i=1}^n y_i} \lambda^{\alpha-1} e^{-\beta\lambda} \\ &\propto \lambda^{\alpha+n-1} e^{-(\sum_{i=1}^n y_i + \beta)\lambda} \end{aligned}$$

Therefore,

$$\lambda \mid y_1, \dots, y_n \sim \text{Gamma}(\alpha + n, \beta + \sum_{i=1}^n y_i)$$

7.

General form of posterior mean and mode can be found on the distribution cheat sheet: - posterior mean:

$$\frac{\alpha+n}{\beta+\sum_{i=1}^n y_i} \text{ - posterior mode: } \frac{\alpha+n-1}{\beta+\sum_{i=1}^n y_i}$$

8.

```
y <- c(2, 30, 40, 10, 24, 14, 6, 8, 20, 35)
```

For the prior, choose $\alpha = 100$ and $\beta = 5$. (Any other reasonable choice will also work.)

```
a <- 100
b <- 5
a_post <- a + length(y)
b_post <- b + sum(y)
# posterior mean #
a_post / b_post
```

```
## [1] 0.5670103
```

```
# posterior mode #
(a_post - 1) / b_post
```

```
## [1] 0.5618557
```

9.

- 95% Central interval: find the 0.025% quantile of $\text{Gamma}(\alpha + n, \beta + \sum_{i=1}^n y_i)$ as the lower bound, and the 0.975% quantile of $\text{Gamma}(\alpha + n, \beta + \sum_{i=1}^n y_i)$ as the upper bound. Can use the **qgamma** function to do this.
- 95% HPD interval: find the interval $C = \{\theta : p(\theta \mid y) \geq k\}$, where k is the largest number such that $\int_{\theta: p(\theta \mid y) \geq k} p(\theta \mid y) d\theta = 1 - \alpha$ and $p(\theta \mid y)$ is the density of $\text{Gamma}(\alpha + n, \beta + \sum_{i=1}^n y_i)$.

Both are regions for which the posterior probability that θ is in the region is equal to 0.95.

10.

- The HPD interval is shorter.
- There is 95% of the chance that λ is within the 95% credible interval.
- For the credible interval, the parameter is random while the endpoints are fixed given the parameter; For the frequentist confidence interval, the parameter is fixed while the endpoints are random.

11.

An alternative to doing calculus is to draw samples from the posterior predictive distribution and approximate the distribution (or an posterior summaries) using Monte Carlo. Pseudocode:

```
for(s in 1:nsamples) {
  1. Sample lambda_s from a Gamma(n + alpha, sum(y) + beta)
  2. Sample ypred_s from an Expo(lambda_s)
}
return ypreds
```

12.

```
for(s in 1:nsamples) {
  1. Sample lambda_s from a Gamma(n + alpha, sum(y) + beta)
  2. Sample ypred_s from an Expo(lambda_s)
}
return ypreds

frac_ytilde_greater <- mean(ypreds > 30)
print(frac_ytilde_greater)
```

##. 13.

The posterior predictive distribution has higher variance than the posterior distribution. It incorporates two sources of variability: posterior variance **and** sampling variance of $\tilde{y}$ given the θ . The addition of the sampling variance of a new observation is what makes the variance larger.

Problem 2

1. Poisson-Gamma Practice

(a)

$$L(\lambda|y_1, y_2, \dots, y_n, \nu_1, \nu_2, \dots, \nu_n) = \prod_{i=1}^n (\nu_i \lambda)^{y_i} e^{-\nu_i \lambda} / y_i! \propto \lambda^{\sum y_i} e^{-\lambda \sum \nu_i}$$

(b)

$$p(\lambda|y) \propto L(\lambda) * p(\lambda) \propto \lambda^{\sum y_i} e^{-\lambda \sum \nu_i} \times \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} \propto \lambda^{\sum y_i + a - 1} e^{-\lambda(\sum \nu_i + b)}$$

So the posterior is a *Gamma*($\sum y_i + a, \sum \nu_i + b$)

(c)

b can be interpreted as the number of prior observations and a can be interpreted as the sum of the counts from prior total exposure of b .

(d)

$$\begin{aligned} E[\lambda \mid y_1, \dots, y_n] &= \frac{a + \sum y_i}{b + \sum \nu_i} \\ &= \frac{b}{b + \sum \nu_i} \frac{a}{b} + \frac{\sum \nu_i}{b + \sum \nu_i} \frac{\sum y_i}{\sum \nu_i} \\ &= (1 - w) \frac{a}{b} + w \hat{\lambda}_{\text{MLE}} \end{aligned}$$

2. Beta-Bernoulli practice

(a)

$$L(\theta) \propto \theta^{\sum y_i} (1 - \theta)^{n - \sum y_i}$$

(b)

$$\begin{aligned} p(\theta) &\propto \theta^{a-1} (1 - \theta)^{b-1} \\ p(\theta \mid y_1, y_2, \dots, y_n) &= \theta^{\sum y_i + a - 1} (1 - \theta)^{n - \sum y_i + b - 1} \end{aligned}$$

So the posterior is a $Beta(\sum y_i + a, n - \sum y_i + b)$

(c)

We can interpret a as prior total number of success and b as prior total number of failures.

(d)

$$E[\theta \mid y_1, y_2, \dots, y_n] = \frac{\sum y_i + a}{n + a + b} = \frac{\sum y_i}{n} \times \frac{n}{n + a + b} + \frac{a}{a + b} \times \frac{a + b}{n + a + b}$$

Multiple Choice Practice

1. a b c
2. a c e
3. c
4. Beta Gamma
5. a
6. c
7. b c
8. a
9. b